

Final examination on Logic Circuit Design

Christian Rinderknecht

13 december 2006

Questions.

1. Let be an n -bit binary number. How many numbers can it encode? What is the lowest? What is the highest? (Prove it.)
2. Same questions if the binary number is interpreted as a 2-complement.
3. When is the number $d_{n-1} \times 10^{n-1} + d_{n-2} \times 10^{n-2} + \dots + d_1 \times 10 + d_0$ interpreted as the decimal number whose digits are $d_{n-1}d_{n-2} \dots d_0$?
4. Prove that, in order to convert a binary number into an octal number, one groups the bits three by three from right to left and then convert each group into decimal, without worrying about carries.
5. Why is the multiplication of binary numbers easy?
6. Consider an n -bit 2-complement binary number. If the leftmost bit is 1, then the number is negative, whilst if it is 0, then the number is positive or nul. Why?
7. Can the negation of an n -bit 2-complement binary number fail?
8. For all bits x and y , the boolean function F is partially defined as

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	1	0	1
0	1	1	1	1
1	1	x	y	0
1	0	0	0	1
1	0	1	1	0

Find both the simplified sum of products and product of sums of F .